Spur Gears Component Generator (Version: 2025.2 (Build 292293000, 293))

5/28/2025

Project Info (iProperties)

Guide

Design Output - Total Unit Correction
Unit Correction Distribution - Custom
Type of Load Calculation - Torque calculation for the specified power and speed
Type of Strength Calculation - Check Calculation
Method of Strength Calculation - ANSI/AGMA 2001-D04:2005

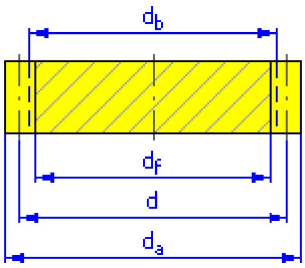
Common Parameters

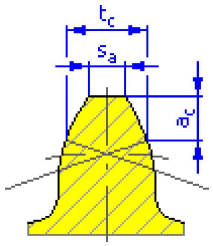
Gear Ratio	i	0.2857 ul
Desired Gear Ratio	i_{in}	0.2857 ul
Module	m	6.000 mm
Helix Angle	β	0.0000 deg
Pressure Angle	α	20.0000 deg
Center Distance	a_w	378.000 mm
Product Center Distance	a	378.000 mm
Total Unit Correction	Σx	0.0000 ul
Circular Pitch	p	18.850 mm
Base Circular Pitch	p_{tb}	17.713 mm
Operating Pressure Angle	α_w	20.0000 deg
Contact Ratio	ε	1.7441 ul
Limit Deviation of Axis Parallelity	f_x	0.0170 mm
Limit Deviation of Axis Parallelity	f_y	0.0085 mm

Gears

		Gear 1	Gear 2
Type of model		Component	Component
Number of Teeth	z	98 ul	28 ul
Unit Correction	x	0.0000 ul	0.0000 ul
Pitch Diameter	d	588.000 mm	168.000 mm
Outside Diameter	d_a	600.000 mm	180.000 mm
Root Diameter	d_f	573.000 mm	153.000 mm
Base Circle Diameter	d_b	552.539 mm	157.868 mm
Work Pitch Diameter	d_w	588.000 mm	168.000 mm
Facewidth	b	72.000 mm	72.000 mm
Facewidth Ratio	b_r	0.1224 ul	0.4286 ul
Addendum	a*	1.0000 ul	1.0000 ul
Clearance	c*	0.2500 ul	0.2500 ul
Root Fillet	r_f*	0.3500 ul	0.3500 ul
Tooth Thickness	s	9.425 mm	9.425 mm

Tangential Tooth Thickness	s_t	9.425 mm	9.425 mm
Chordal Thickness	t_c	8.322 mm	8.322 mm
Chordal Addendum	a_c	4.485 mm	4.485 mm
Chordal Dimension	W	211.932 mm	64.348 mm
Chordal Dimension Teeth	z_w	12.000 ul	4.000 ul
Dimension Over (Between) Wires	M	605.041 mm	184.627 mm
Wire Diameter	d_M	11.000 mm	11.000 mm
Limit Deviation of Helix Angle	F_β	0.0170 mm	0.0150 mm
Limit Circumferential Run-out	F_r	0.0490 mm	0.0300 mm
Limit Deviation of Axial Pitch	f_{pt}	0.0140 mm	0.0110 mm
Limit Deviation of Basic Pitch	f_{pb}	0.0130 mm	0.0100 mm
Virtual Number of Teeth	z_v	98.000 ul	28.000 ul
Virtual Pitch Diameter	d_n	588.000 mm	168.000 mm
Virtual Outside Diameter	d_{an}	600.000 mm	180.000 mm
Virtual Base Circle Diameter	d_{bn}	552.539 mm	157.868 mm
Unit Correction without Tapering	x_z	-1.9133 ul	0.2086 ul
Unit Correction without Undercut	x_p	-4.7122 ul	-0.6180 ul
Unit Correction Allowed Undercut	x_d	-4.8822 ul	-0.7879 ul
Addendum Truncation	k	0.0000 ul	0.0000 ul
Unit Outside Tooth Thickness	s_a	0.8065 ul	0.7310 ul
Tip Pressure Angle	α_a	22.9422 deg	28.7119 deg





Loads

		Gear 1	Gear 2
Power	P	55.000 kW	55.000 kW
Speed	n	427.50 rpm	1496.25 rpm
Torque	T	1228.564 N m	351.018 N m
Efficiency	η	1.000 ul	
Radial Force	F_r	1520.955 N	
Tangential Force	F_t	4178.791 N	
Axial Force	F_a	0.000 N	
Normal Force	F_n	4446.976 N	
Circumferential Speed	v	13.162 mps	
Resonance Speed	n_{E1}	1506.036 rpm	

Material

	Gear 1	Gear 2
	AISI 4140	AISI 4140

Ultimate Tensile Strength	S_u	1130 MPa	1130 MPa
Yield Strength	S_y	885 MPa	885 MPa
Modulus of Elasticity	E	195000 MPa	195000 MPa
Poisson's Ratio	μ	0.290 ul	0.290 ul
Allowable Bending Stress	s_{at}	413.9 MPa	404.6 MPa
Allowable Contact Stress	s_{ac}	1485.8 MPa	1485.8 MPa
Hardness in Tooth Core	JHV	210 ul	210 ul
Type of Treatment	type	4 ul	4 ul

Strength Calculation

Factors of Additional Load

Overload Factor	K_o	1.000 ul	
Dynamic Factor	K_v	1.116 ul	
Size Factor	K_s	1.000 ul	1.000 ul
Reliability Factor	K_R	0.850 ul	
Temperature Factor	k_t	1.000 ul	
Load Distribution Factor	K_m	1.224 ul	1.224 ul
Lead Correction Factor	C_{mc}	1.000 ul	1.000 ul
Mesh Alignment Correction Factor	C_e	1.000 ul	
Pinion Proportion Modifier	C_{pm}	1.100 ul	
Mesh Alignment Factor	C_{ma}	Commercial Enclosed Gear Units (0.1709)	

Factors for Contact

Surface Condition Factor	C_f	1.000 ul	1.000 ul
Stress Cycle Factor	Z_N	0.834 ul	0.777 ul
Hardness Ratio Factor	C_H	1.000 ul	1.000 ul
Elastic Factor	C_p	2216.888 ul	
Geometry Factor	I	0.115 ul	

Factors for Bending

Reverse Loading Factor	Y_a	1.000 ul	1.000 ul
Rim Thickness Factor	K_B	1.000 ul	1.000 ul
Stress Cycle Factor	Y_N	0.901 ul	0.865 ul
Geometry Factor	J	0.612 ul	0.599 ul

Results

Factor of Safety from Pitting	k_f	3.902 ul	3.637 ul
Factor of Safety from Tooth Breakage	k_n	20.309 ul	18.652 ul
Check Calculation		**Positive**	

Summary of Messages

5:52:47 AM Design: Numbers of teeth are commensurable - shots of the same teeth are taken relatively regularly
5:52:47 AM Design: Gear 2: The Unit Correction (x) is less than the Unit Correction without Tapering (x_z)

5:52:47 AM Calculation: Calculation indicates design compliance!